APIs and Election Forecasting

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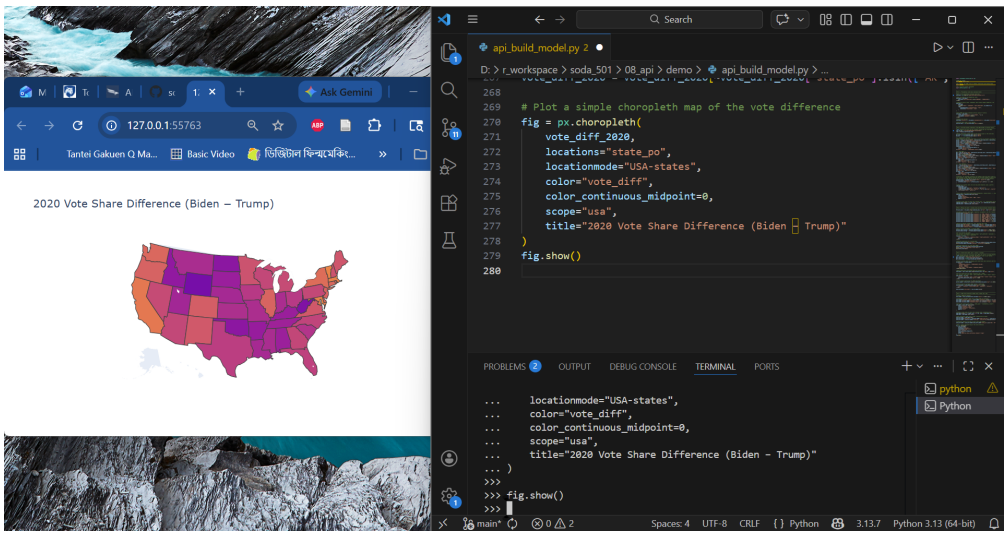
Start Off: Verifying Your Environment

1. **Environment check (required).** Submit proof that you successfully ran the full tutorial code on your machine. You may submit *one* of the following:
   - A screenshot or text file showing console output that includes the printed representation shapes or output.



Conceptual Questions

Please write three to ten sentence explanations for each of the following questions. **You are only required to answer ONE of the two questions below.**

2. In the social sciences, what are two advantages of collecting data via an API (instead of web scraping)? What are two limitations or risks (e.g., coverage bias, changing endpoints, rate limits, versioning, missingness)? Explain how you would document API-based data provenance so another researcher could replicate your dataset later.

   Response: APIs provide structured, machine-readable responses that are more stable and reproducible than scraping HTML, which breaks when page layouts change. They also reflect the data provider's intended access interface, meaning the fields returned are predictable and consistently formatted. On the other hand, rate limits constrain how much data you can collect within a given time window, directly shaping what research questions are feasible.

APIs also embed coverage decisions - if a field isn't exposed or an endpoint is deprecated, that data simply doesn't exist, introducing silent missingness. To document provenance for replication, we need to record the endpoint URL, query parameters, authentication type, collection date, and API version so another researcher can reconstruct exactly what was requested and when.

Applied Exercises

Use the code in the week's code tutorial and the lecture slides to answer the following questions.

Response: Find this week's work here: `https://github.com/yasinshafi/soda501/tree/main/08_api`

4. **Create an API key and connect to the API.** This week you will use the `fredr` package, which requires a FRED API key.
   - Create a FRED API key.
   - Update the provided script so it reads your key from a local environment variable (recommended) rather than hard-coding it in the file.
   - Confirm your API call works by successfully downloading at least one FRED time series (e.g., unemployment, GDP, or CPI) for election years.
5. **Run the baseline forecaster (replicate).** Using the provided code (`api_build_model.r`), get the baseline pipeline running end-to-end on your machine.
   - If any file paths are absolute, modify them to use a **relative path** that will work inside your course project folder.
   - Produce at least one baseline output showing model predictions on held-out data (e.g., 2020), such as a printed table, summary statistics, or a map.
6. **Build a better out-of-sample forecaster.** Starting from the baseline, improve out-of-sample performance for election results that are not used to fit the model.
   - Define an explicit out-of-sample design (choose one):
     (a) Hold out the most recent election year in the dataset (e.g., train on years < 2020, test on 2020), *or*
     (b) Do a "leave-one-election-out" evaluation (train on all but one election year; repeat).
   - Implement **at least two** model improvements. Examples include:
     - adding additional FRED indicators (via the API) and justifying them,
     - changing the functional form (interactions, nonlinear terms),
     - using regularization (ridge/lasso/elastic net),
     - switching model families (e.g., random forest, gradient boosting),
     - improving feature engineering (e.g., transforms, per-capita changes),
     - improving how you aggregate to a prediction target (national vs state).
   - Report performance **out-of-sample** using at least one quantitative metric (e.g., MAE, RMSE) and compare it to the baseline.
7. **Communicate the comparison.** Provide:
   - one table that compares baseline vs improved performance (out-of-sample), and
   - one figure that communicates model fit or errors (e.g., predicted vs actual; error by state; time series comparison).

Challenge Question (Optional — if you finish early)

7. **Forecast validation and "small n" in elections.** Defend a position (8–12 sentences): Do we have enough realized election outcomes to validate a national election forecaster? Discuss the trade-offs of treating forecasting as (i) national outcomes (one datapoint per election) versus (ii) state-level outcomes (many datapoints per election). Explain what you would report as evidence that your model is "working" and what would still remain uncertain.
8. **API engineering + reproducibility challenge.** Write a small wrapper function `get_fred_features(seri years)` that:
   - downloads multiple FRED series via the API,
   - caches the raw results to disk (so repeated runs do not re-download),
   - gracefully handles API failures (e.g., `tryCatch` + informative messages).

   Demonstrate it by adding **one new** economic series to your model and showing whether it helps out-of-sample performance.